Topic 6D: Halogenoalkanes
30. know that halogenoalkanes can be classified as primary, secondary or tertiary
31. understand what is meant by the term ‘nucleophile’
32. understand the reactions of halogenoalkanes with: i aqueous potassium hydroxide to produce alcohols (where the hydroxide ion acts as a nucleophile) ii aqueous silver nitrate in ethanol (where water acts as a nucleophile) iii potassium cyanide to produce nitriles (where the cyanide ion acts as a nucleophile) *Students should know this as an example of increasing the length of the carbon chain.* iv ammonia to produce primary amines (where the ammonia molecule acts as a nucleophile) v ethanolic potassium hydroxide to produce alkenes (where the hydroxide ion acts as a base)

Students should:
33. understand that experimental observations and data can be used to compare the relative rates of hydrolysis of: i primary, secondary and tertiary halogenoalkanes ii chloro-, bromo-, and iodoalkanes using aqueous silver nitrate in ethanol
CORE PRACTICAL 4: Investigation of the rates of hydrolysis of some halogenoalkanes
34. know the trend in reactivity of primary, secondary and tertiary halogenoalkanes
35. understand, in terms of bond enthalpy, the trend in reactivity of chloro-, bromo-, and iodoalkanes
36. understand the mechanisms of the nucleophilic substitution reactions between primary halogenoalkanes and: i aqueous potassium hydroxide ii ammonia